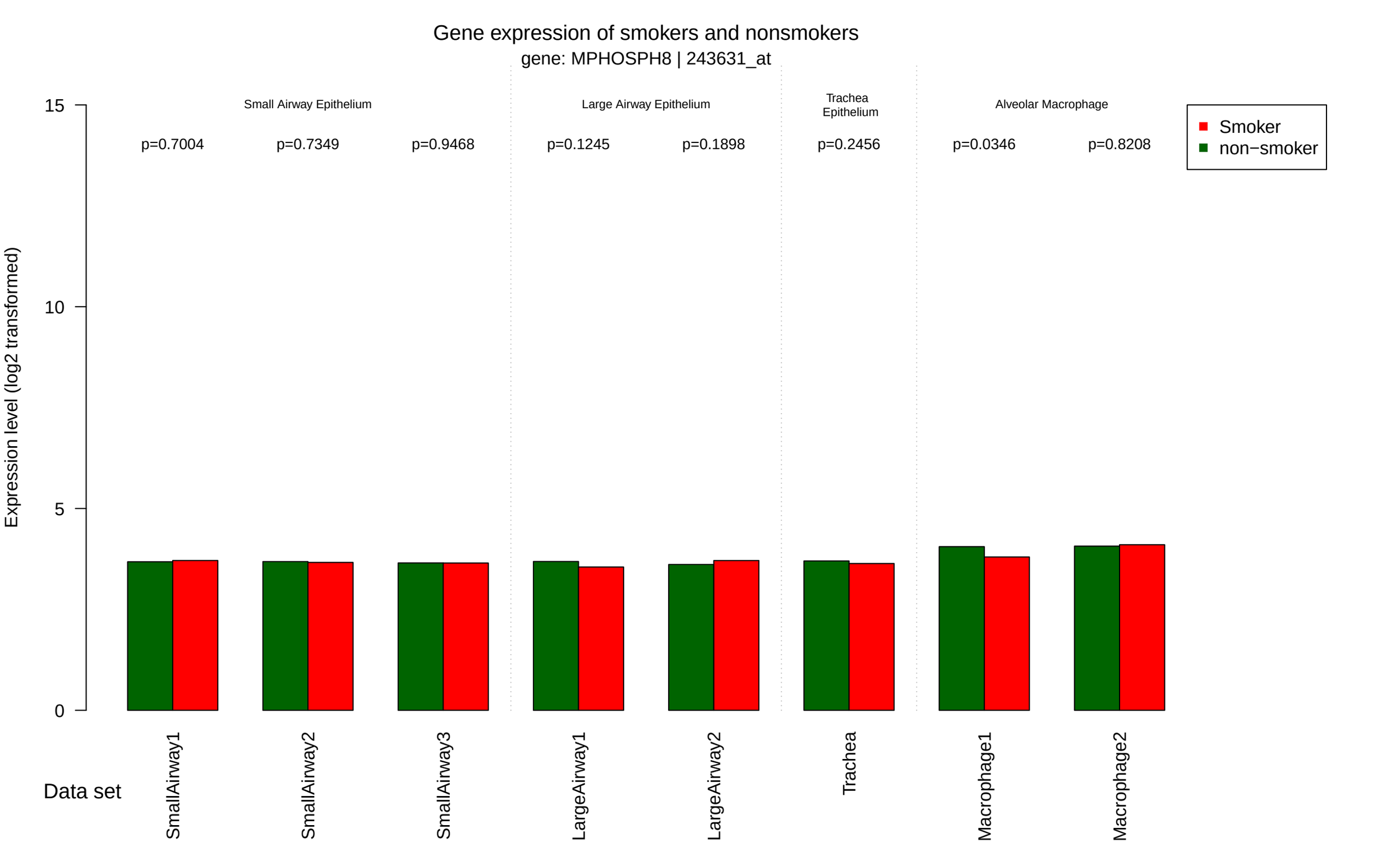
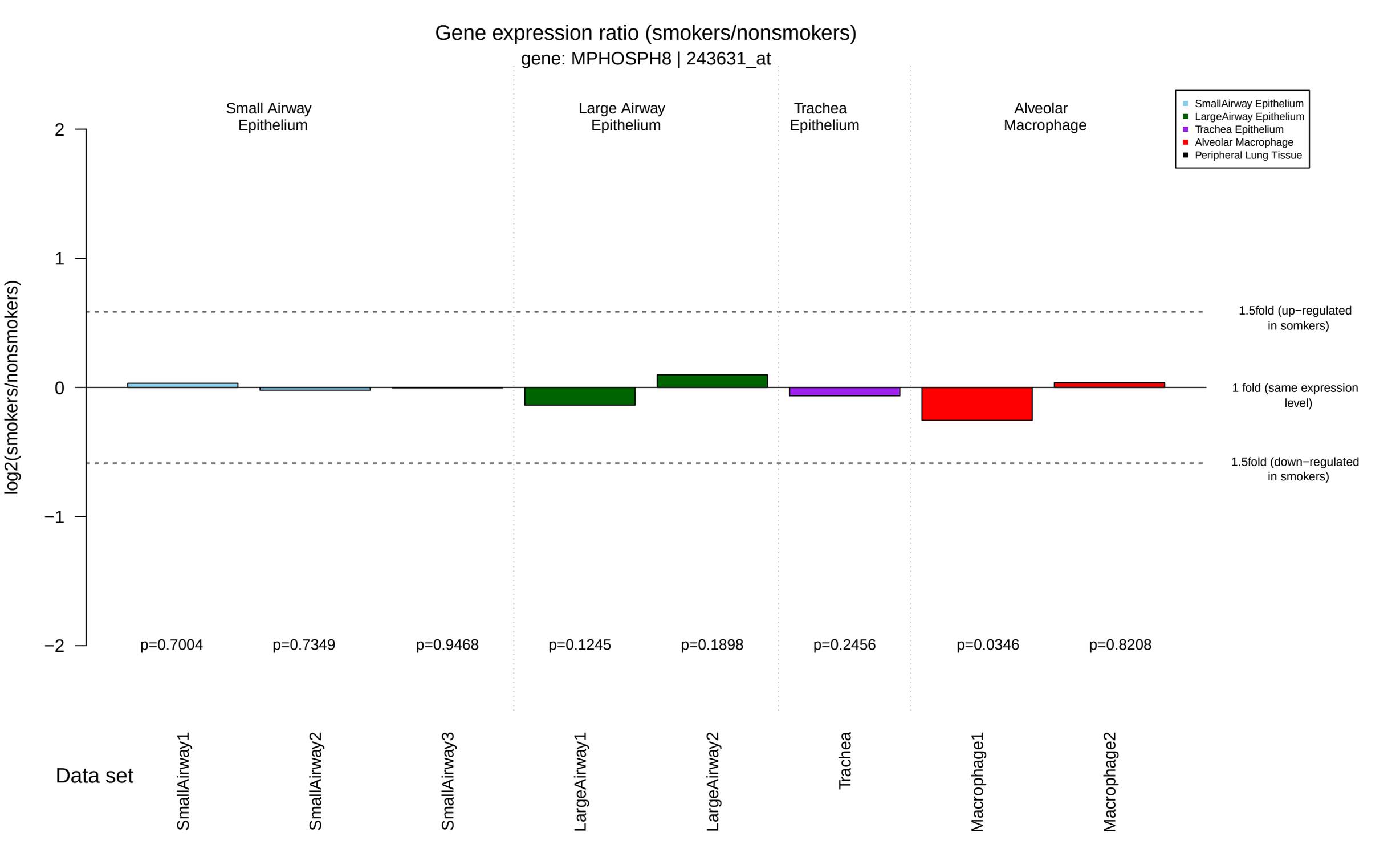
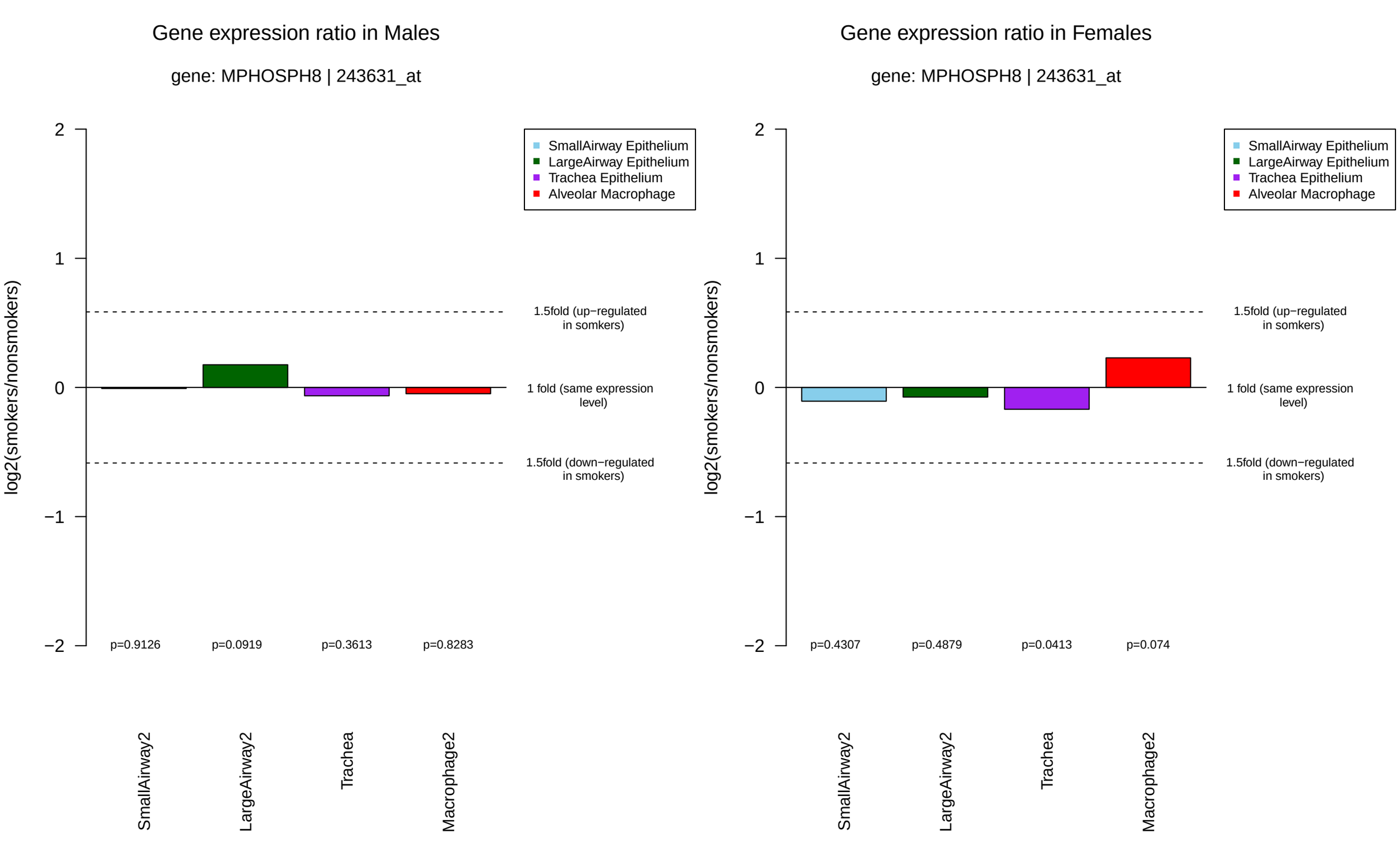
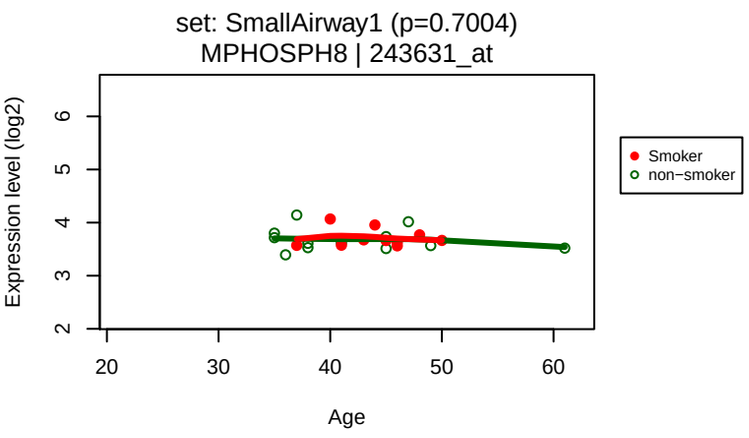
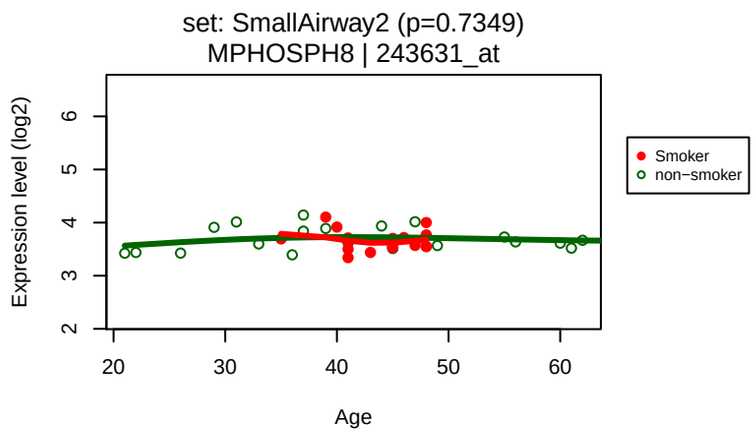
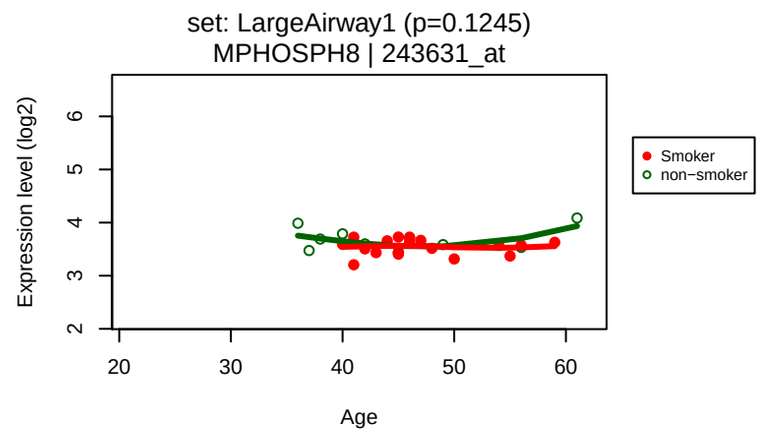

Gene expression of smokers and nonsmokers
gene: MPHOSPH8 | 243631_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7004
p=0.7349
p=0.9468
p=0.1245
p=0.1898
p=0.2456
p=0.0346
p=0.8208
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

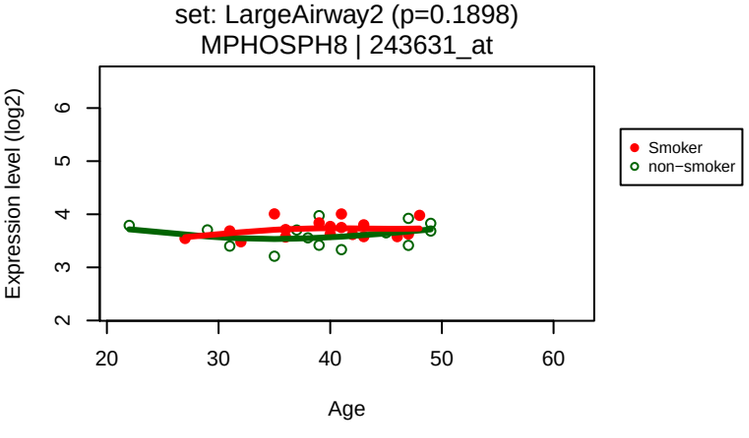

Gene expression ratio (smokers/nonsmokers)
gene: MPHOSPH8 | 243631_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7004
p=0.7349
p=0.9468
p=0.1245
p=0.1898
p=0.2456
p=0.0346
p=0.8208
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

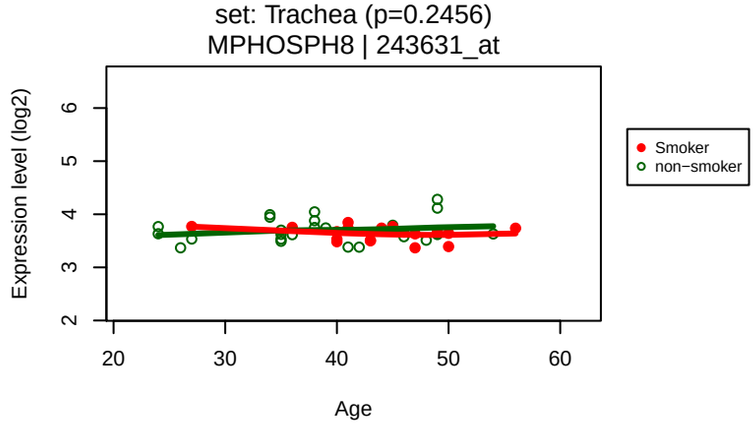

Gene expression ratio in Males
gene: MPHOSPH8 | 243631_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9126
p=0.0919
p=0.3613
p=0.8283
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MPHOSPH8 | 243631_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4307
p=0.4879
p=0.0413
p=0.074
SmallAirway2
LargeAirway2
Trachea
Macrophage2

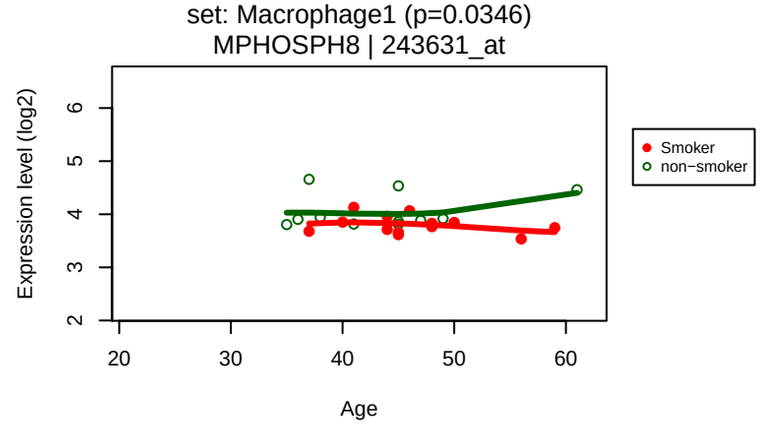

set: SmallAirway1 (p=0.7004)
MPHOSPH8 | 243631_at
Expression level (log2)
Age
Smoker
non−smoker

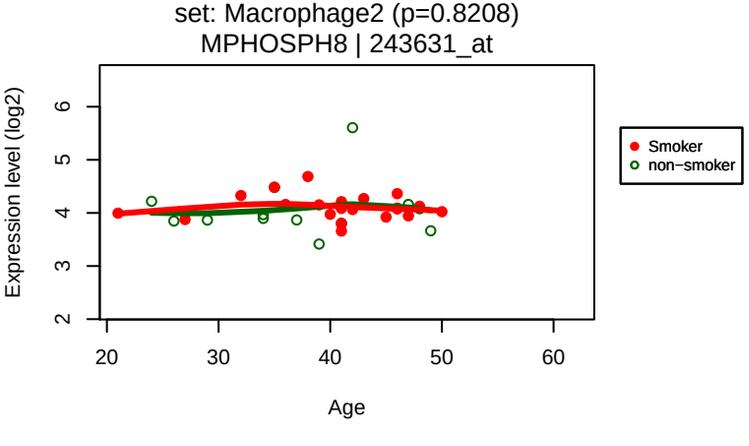

set: SmallAirway2 (p=0.7349)
MPHOSPH8 | 243631_at
Expression level (log2)
Age
Smoker
non−smoker

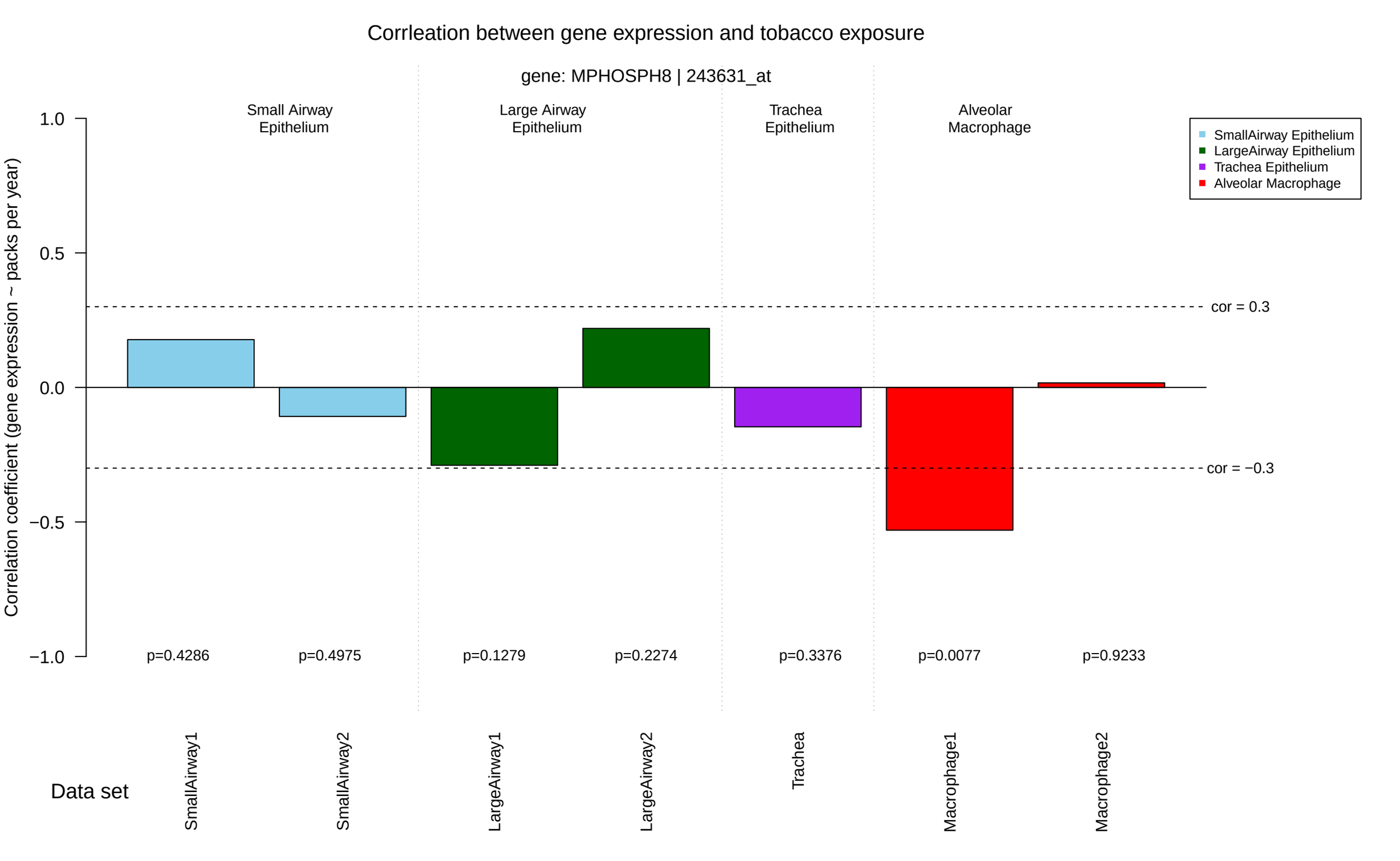

set: LargeAirway1 (p=0.1245)
MPHOSPH8 | 243631_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1898)
MPHOSPH8 | 243631_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2456)
MPHOSPH8 | 243631_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0346)
MPHOSPH8 | 243631_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8208)
MPHOSPH8 | 243631_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MPHOSPH8 | 243631_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4286
p=0.4975
p=0.1279
p=0.2274
p=0.3376
p=0.0077
p=0.9233
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2